

- 2 The cubic equation $x^3 + 4x^2 + 6x + 1 = 0$ has roots α, β, γ .

(a) Find the value of $\alpha^2 + \beta^2 + \gamma^2$. [2]

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(b) Use standard results from the list of formulae (MF19) to show that

$$\sum_{r=1}^n ((\alpha + r)^2 + (\beta + r)^2 + (\gamma + r)^2) = n(n^2 + an + b),$$

where a and b are constants to be determined. [6]

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